

Franco Terranova

Ph.D. Candidate, Université de Lorraine / INRIA, Nancy, France
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EDUCATION

Université de Lorraine & INRIA, Nancy, France 10/2023 — 10/2026
Ph.D. in Computer Science - RESIST Team / SuperviZ Project (France 2030)

- Design of a sequential decision-making framework for identifying critical vulnerability paths in networks
- Design of invariant and transferable embedding action spaces for scalable and generalizable RL solutions
- Extension of Microsoft CyberBattleSim with a more realistic cyber-attack simulation (C-CyberBattleSim)
- Developed a LM-based framework for vulnerability analysis and dynamics estimation (CVE-LMTune)

University of Pisa, Pisa, Italy 09/2021 — 09/2023
Master's Degree in Artificial Intelligence and Data Engineering Final Grade: (110/110) cum laude

University of Pisa, Pisa, Italy 09/2018 — 07/2021
Bachelor's Degree in Computer Engineering Final Grade: (110/110) cum laude

EXPERIENCE

Universitat Politècnica de Catalunya Barcelona, Spain
Visiting Researcher - Barcelona Neural Networking Center 11/2025 — 03/2026
Project: GNN-based embedding spaces for boosting RL heuristics in graph combinatorial optimization

University of Waterloo Waterloo, Canada
Visiting Researcher - Cheriton School of Computer Science 06/2024 — 12/2024
Project: Multi-agent RL for a secure and efficient virtual machine placement

Fermi National Accelerator Laboratory Chicago, Illinois
Master Degree Thesis Researcher - Deep Skies Lab 05/2023 — 08/2023
Project: Deep RL for autonomous telescope scheduling

European Space Agency Köln, Germany
AI/ML Engineer Intern - Spaceship EAC Team 11/2022 - 04/2023
Project: Deep learning for real-time detection of spaceflight-related ocular conditions

Fermi National Accelerator Laboratory Chicago, Illinois
Summer Intern - GlidenWMS Project 07/2022 — 09/2022
Project: Flexible pilot jobs framework for distributed high throughput computing

ACADEMIC ENGAGEMENT

Teaching Assistant
Data Science IDMC, Nancy, France - 03/2026
Introduction to Natural Language Processing IDMC, Nancy, France - 09/2025
Project Management Tools IDMC, Nancy, France - 09/2025
Probability and Statistics IDMC, Nancy, France - 09/2025
Data Science IDMC, Nancy, France - 03/2025
Méthodes et Outils pour l'Analyse du Comportement Humain IDMC, Nancy, France - 01/2025
Symbolic Artificial Intelligence IDMC, Nancy, France - 01/2025

Tutorials
Tutorial From RL to Meta-RL European Agent Systems Summer School, Romania - 09/2025
Introduction to Reinforcement Learning & Deep RL DeepLorIA, France - 11/2024
Foundations of Deep RL and Environment Setup European Agent Systems Summer School, Ireland - 08/2024
Organizer of the DeepLorIA network: delivered 10+ tutorials on AI topics LORIA, Nancy, France - 2024–2026

Talks & Seminars
A More Realistic Simulation for AI-driven Attack Path Discovery Black Hat Asia 2026, Singapore - 04/2026
Sélection de Paradigmes et Validation Statistique: Bonnes Pratiques en ML University of Lorraine, France - 12/2025

Modeling Attacker Behavior with RL Agents to Anticipate Vulnerability Paths	University of Tokyo, Japan - 11/2025
Learning to Predict Cyber Attack Paths with Reinforcement Learning	Cybersecurity Workshop, Tokyo, Japan - 11/2025
Can AI Help Us Understand Vulnerabilities?	Forum InCyber Europe, Lille, France - 03/2025
Discovering Critical Vulnerability Paths using RL Agents	Campus Cyber, Paris, France - 03/2025
Deep RL for Cyber-Attack Path Prediction	INRIA Rennes, France - 03/2024
Autonomous Telescope Scheduling with Reinforcement Learning	PyHEP 2023 Workshop, Online - 10/2023
Introduction to Reinforcement Learning	Fermilab Cross-lab AI Meetings, Chicago, Illinois - 07/2023

Summer Schools

OxML Summer School on Representation Learning	Oxford, UK - 08/2025
Geilo Winter School on Graphs & Applications	Geilo, Norway - 01/2024
Fermilab Summer School	Chicago, Illinois - 07/2022

RESEARCH

Selected Conference Papers

- Terranova, F., Rekbi, S., Lahmadi, A., Chrisment, I. "Multi-Taxonomy Vulnerability Classification with Hierarchically Fine-Tuned Language Models". To appear at the International Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2026).
- Terranova, F., Lahmadi, A. & Chrisment, I. (2025). "Scalable and Generalizable RL Agents for Attack Path Discovery via Continuous Invariant Spaces," in Proceedings of the 28th International Symposium on Research in Attacks, Intrusions and Defenses (RAID'25).
- Terranova, F., Lahmadi, A. & Chrisment, I. (2024). "Leveraging Deep Reinforcement Learning for Cyber-Attack Paths Prediction: Formulation, Generalization, and Evaluation," in Proceedings of the 27th International Symposium on Research in Attacks, Intrusions and Defenses (RAID'24).
- Terranova, F., Voetberg, M., Nord, B. & Pagul, A. (2023). "Self-Driving Telescopes: Autonomous Scheduling of Astronomical Observation Campaigns with Offline Reinforcement Learning," in Machine Learning and the Physical Sciences Workshop, NeurIPS 2023.

Patent: Ritter, S., Drescher, J., Stern, C., Terranova, F., Cowley, A. (2025). "Mobile Device and Computer-Implemented Method for Real-Time Retinal Diagnosis, Data Processing Apparatus, Computer Program, and Computer-Readable Medium," filed in US and EU patent offices.

Peer Reviewing: IJCNN 2026, NOMS 2026, NeurIPS 2025 (Position Paper Track), IJCNN 2025, AAAI 2024 (Workshop), Computational and Structural Biotechnology Journal (2024), RESSI 2025

Supervision: 1 IDMC M2 Master Thesis, 1 IDMC M1 Summer Intern, 4 IDMC M1 Semester Students & 1 University of Cincinnati Intern on the project "Automated Analysis of Vulnerabilities Using NLP" (2025)

EXTRA

Mentoring: Mentee for the Université de Lorraine Mentorship Program (2024), LeadTheFuture Mentorship Program (2023), Young ISSNAF Mentoring Program for Students (2023), and SGAC Mentorship Program (2023)

Memberships: Member of the AI Doctoral Academy (since 2023), IEEE, IEEE Computer Society, IEEE Computational Intelligence Society & IEEE Communications Society Member (since 2024), ACM, and ACM SIGAI Member (since 2024)

Social Engagement: AI-enabled Proactive Cybersecurity Stand, Nuit de la Science & Fête de la Science, Nancy, France (2025); Scrutineer for the European Elections, Nancy, France (2024); Booth Staff for the European Space Agency at Festival Yggdrasil, Eurexpo Lyon, France (2023)

OTHER PROJECTS

Internet of Things for a Smart Regolith Transportation System	Golden, Colorado
<i>Over The Dusty Moon Challenge 2023</i> - Colorado School of Mines / Lockheed Martin	12/2022 — 06/2023

Cyber-Attack Response with Graph Neural Networks and Language Models	Milan, Italy
<i>European Union Defense Innovation Scheme Hackaton 2024</i>	05/2024 — 06/2024

LANGUAGES

Italian: Native **English:** C1/C2 (Professional proficiency) **French:** C1 (Advanced) **Spanish:** A2 (Basic)